Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Therapeutic Technologies for Modulating Mitochondrial Dysfunction and Addressing Expanded Repeat Pathology in Repeat Expansion Diseases			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio·Medical		Bio·Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Gene Pharmaceuticals	Medical Bio	Small-Molecule Pharmaceuticals
Concept of Proposed Technology	Mitochondrial dysfunction and impaired cellular energy metabolism are increasingly recognized as core pathological drivers across a wide spectrum of rare, neurological, neuromuscular, and metabolic diseases[1]. Among them, repeat expansion disorders (REDs) such as Huntington's disease, Friedreich's ataxia, C9orf72-associated ALS/FTD, Fragile X syndrome, and myotonic dystrophy type 1 share a convergent vulnerability in mitochondrial dysfunction and impaired cellular energy metabolism, which increasingly appear to be central drivers of pathogenesis rather than secondary consequences[2-4]. These disorders arise when short DNA tandem repeats, which are normally present, expand due to unknown mechanisms. Expanded repeat tracts disrupt DNA, RNA, and protein homeostasis; toxic RNA structures, RAN-translated peptides, and disrupted repair pathways converge to impair mitochondrial signaling, redox regulation, and calcium homeostasis, ultimately compromising ATP production and bioenergetic capacity. Recent comprehensive analyses demonstrate that REDs exhibit defects in mitochondrial respiration, increased ROS			

	generation, impaired mitochondrial dynamics and biogenesis, and dysregulated mitophagy, collectively leading to metabolic stress that exacerbates neuronal and muscular degeneration. These mitochondrial abnormalities are consistently observed across distinct REDs, highlighting shared pathogenic nodes and suggesting that impaired cellular energy metabolism is a unifying mechanistic axis underlying diverse clinical phenotypes. Dysregulation of pathways related to ATP production, redox balance, and NAD^+ homeostasis can lead to progressive cellular damage, oxidative stress, and tissue degeneration, particularly in energy-demanding organs such as the brain, heart, retina and skeletal muscle[5]. Recent advances in therapy have seen successful development of drugs targeting mitochondrial diseases, validating proof-of-concept for mitochondrial-directed strategies that restore mitochondrial and cellular bioenergetic function[6-8]. Nevertheless, approaches that can eliminates the root cause, as well as the broader applicability of mitochondrial targeting strategies across complex neurological and metabolic disease states remains to be fully established. The objective of this joint R&D program is to support the development of first-in-class or best-in-class therapeutic concepts that: • Address mitochondrial dysfunction–associated repeat expansion diseases by targeting the underlying causes and restoring mitochondrial function and cellular energy metabolism through novel biological targets or pathways; • Go beyond symptomatic relief or indirect stress-response activation by addressing disease-driving pathogenic bottlenecks; • Are applicable to rare diseases and neurological disorders where unmet medical need remains high. Proposed technologies may include, but are not limited to: • Discovery and validation of new molecular targets involved in mitochondrial repeat expansion diseases; • Development of small molecules or other therapeutic modalities that address the causal factors of the REDs and/or rescue mitochondrial disease phenotypes – e.g., enhance NAD^+ or ATP generation, restore redox balance, or improve mitochondrial resilience; • Platform technologies with potential indication expansion across multiple mitochondrial, neurological, and/or metabolism-related diseases and other REDs.
Rationale for Joint R&D Project	Diseases driven by expanded repeats and mitochondrial/metabolic dysfunction often present complex pathophysiology, limited patient populations, and insufficient disease models, making them challenging for any single organization to address independently. This RFP is based on the following collaboration rationale: • Korean startups/SMEs possess strong capabilities in compound discovery, platform technology development, and early-stage translational research, but often face limitations in advanced disease modeling and *in vivo* validation.

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	• U.S. universities and research institutes offer internationally recognized strengths in disease biology, rare disease models, mechanistic research, and advanced analytical technologies. By combining these complementary strengths, Korea–US joint teams can: • Accelerate the generation of robust preclinical proof-of-concept (PoC); • De-risk innovative therapeutic strategies through rigorous mechanistic and in vivo validation; • Establish globally competitive pipelines aligned with future clinical development and partnering opportunities.
Key Words	1. Repeat expansion disorders 2. Mitochondrial dysfunction 3. NAD^+ 4. ATP

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	Proposals submitted in response to this RFP should aim to achieve the following high-level objectives: 1. Identification and validation of novel therapeutic targets or mechanisms related to mitochondrial or cellular energy dysfunction in repeat expansion diseases. 2. Demonstration of therapeutic efficacy in relevant cellular and/or animal disease models. 3. Mechanistic understanding of how the proposed intervention modulates energy metabolism, redox balance, or mitochondrial function. 4. Mechanistic insight into the processes by which expanded repeats can be actively addressed. 5. Selection of preclinical candidate(s) or lead technologies suitable for further development. Applicants are encouraged to design projects that advance technology readiness from early exploratory research (TRL 2–3) toward preclinical validation (TRL 4–5).			
R&D Target Specifications	• Mitochondrial function and energy metabolism - Increase in ATP production: ≥30% increase versus disease controls - Improvement in mitochondrial respiration (OCR): Achieve a ≥20% increase in maximal respiration versus disease controls - Reduction in ROS: ≥20% reduction in total ROS versus disease controls - Increase in NAD^+/NADH ratio: ≥30% increase versus disease controls • Expanded-repeat pathology - Reduction of expanded-repeat burden: Achieve a ≥20% reduction in quantitative expanded-repeat metrics (e.g., decrease in epigenetic dysregulation, mean/median repeat-length distribution, and/or reduction in the proportion of cells above the pathogenic cut-off) • Disease-relevant functional recovery - In vivo functional improvement: In relevant animal models, achieve a ≥20% improvement in at least one key functional endpoint (e.g., rotarod latency, grip strength, gait metrics) versus disease control • Developability - Target-tissue exposure (e.g., CNS, muscle, heart as appropriate): achieves Kp ≥ 0.1 at efficacious dosing - Early genetic toxicology and cardiac safety a) AMES (bacterial reverse mutation): Negative b) hERG inhibition (in vitro): IC_{50} ≥ 10 μM c) In vitro micronucleus or chromosome aberration assay: Negative - Exploratory safety and tolerability (7 day repeat-dose exploratory study, non-GLP) a) No mortality and no severe clinical sings b) ≤10% body-weight loss c) No clinically meaningful changes in key clinical chemistry/hematology panels			

	(liver/kidney/hematology) • Preclinical Lead Candidates: Discovery of therapeutic candidates and completion of preliminary toxicity studies to support future GLP-compliant preclinical development.
R&D Plan	While detailed experimental plans should be defined by applicant teams, the anticipated scope of R&D may include: • In vitro evaluation of mitochondrial function, ATP production, oxidative stress, and metabolic parameters; • In vitro evaluation of changes in repeat lengths and/or their secondary effects in DNA/RNA/Protein levels • Establishment or utilization of disease-relevant *in vivo* models in collaboration with U.S. research partners; • Pharmacokinetic, ADME, or feasibility studies, as appropriate to the proposed modality; • Exploratory safety or tolerability assessments to support preclinical decision-making. A high-level project timeline of approximately 2–3 years may be proposed; however, specific milestones, methodologies, and task allocation must be justified by the applicants.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Development and application of advanced disease models (cellular and/or animal); • In-depth mechanistic studies and biomarker analysis; • Publication of high-impact scientific findings.
Expected R&R of Korean SME Partner	• Discovery or optimization of therapeutic candidates or platform technologies; • In vitro screening, translational assays, and feasibility studies; • Intellectual property generation and future development planning.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected Technological Outcomes** • Validation of innovative therapeutic concepts addressing mitochondrial or metabolic dysfunction; • Generation of preclinical data suitable for follow-on funding, licensing, or clinical translation; • Establishment of durable Korea–US research partnerships. **Expected Commercial Outcomes** • Creation of globally competitive pipelines in rare and neurological disease therapeutics; • Enhanced commercialization potential through licensing or co-development; • Contribution to improved treatment options for patients with high unmet medical need.

6. References

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